

AMAN LUTHRA

Noida, Uttar Pradesh | +91-9720622558 | amanluthra804@gmail.com

[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Computer Science undergraduate with strong foundations in Data Structures & Algorithms, Object-Oriented Design, Operating Systems, and complexity analysis, proven through 500+ solved problems (LeetCode 1800+, CodeChef 3-Star). Hands-on experience building backend systems, relational databases, and scalable infrastructure pipelines. Comfortable working through broadly defined, ambiguous problems and designing solutions from first principles.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida | B.Tech, Computer Science & Engineering | CGPA: **9.16/10**

2023 – 2027

Asha Modern School | Class XII: **94.4%** | Class X: **92.2%**

TECHNICAL SKILLS

Languages: C++, Python, SQL

Data Structures & Algorithms: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Dynamic Programming, Recursion, Sorting & Searching, Greedy, Backtracking, Hashing

CS Fundamentals: Object-Oriented Design, Operating Systems, Computer Networks, DBMS, Complexity Analysis

Backend & Databases: REST APIs, MySQL (Relational Databases), MongoDB

Systems & Tools: Docker, Jenkins CI/CD, Linux, Git, GitHub

EXPERIENCE

Software Development Intern | US4 Business Solutions Pvt. Ltd., Gurgaon

June 2026 – July 2026

- Designed and developed backend REST APIs to fetch, process, and serve real-time health metrics (heart rate, SpO₂, sleep, anxiety) from wearable devices.
- Applied object-oriented design principles while integrating wearable data sources into the application backend.
- Collaborated with the AI/ML team, structuring clean datasets to support automated analysis and downstream systems.
- Assisted in debugging, testing, and API integration, working through ambiguous, broadly defined problems independently.

PROJECTS

Warehouse-to-Store Spare Parts Distribution System (C++) | [GitHub](#) | Jul 2025 – Oct 2025

Tech Stack: C++, OOP, Dijkstra's Algorithm, Kruskal's MST

- Designed a distribution optimization system in C++ applying Dijkstra's algorithm for shortest-path routing and Kruskal's MST to minimize total distribution cost across a multi-node network.
- Applied object-oriented design and complexity analysis to build modular, efficient components for inventory tracking and route allocation.
- Compiled and tested entirely within a Linux (Ubuntu) environment using Git-based version control.

DevOps CI/CD Deployment Platform | [GitHub](#) | Mar 2026 – May 2026

Tech Stack: Docker, Docker Compose, Jenkins CI/CD, Nginx, Prometheus, Linux

- Built a containerized, scalable deployment pipeline using Docker and Jenkins CI/CD, automating build, test, and deployment across multiple services.
- Configured Nginx and Prometheus for reverse proxying and real-time monitoring, improving fault tolerance and system observability.
- Designed the pipeline for consistent, reproducible deployments, reducing manual effort on every code push.

PregnaCare — Maternal & Fetal Health Prediction System | [GitHub](#) | Jan 2026 – Apr 2026

Tech Stack: Python (Flask), React.js, MySQL, MongoDB, JWT, RBAC

- Designed and built a full-stack platform with a Flask REST API backend and a relational (MySQL) database, applying OOP principles for a modular, maintainable codebase.
- Implemented JWT-based authentication and role-based access control (RBAC), designing the schema for efficient queries at scale.

CERTIFICATIONS & ACHIEVEMENTS

Innovate 2.0: Hack to Build — Round 2 Qualifier, Online Hackathon (Unstop) | [Certificate](#)

500+ DSA Problems, LeetCode 1800+ — strong analytical and problem-solving ability | [Profile](#)

Goldman Sachs SWE Job Simulation — Forage (cryptography, secure engineering practices) | [Certificate](#)

JavaScript Algorithms & Data Structures — freeCodeCamp Certification | [Certificate](#)

Python — CIQ & UniAthena | [Certificate](#) Data Science — CIQ & UniAthena | [Certificate](#)

GoFr Summer of Code — Open-Source Participant (Unstop) | [Certificate](#)

Software Development Internship Certificate — US4 Business Solutions | [Certificate](#)